

Business Problem Statement

Indian Retail E-commerce Performance, Customer Experience and Order Fulfillment Optimization

Prepared from the Indian Retail E-commerce dataset | 100,250 transaction records | Analysis period: 2023–2024

The business needs a data-driven way to grow profitable sales while reducing revenue leakage from cancellations and returns, improving fulfillment visibility, and identifying the customer and product segments with the greatest opportunity.

1. Business context

The dataset represents a multi-category Indian retail e-commerce operation covering customer demographics, geography, product categories, order economics, discounts, payment methods, sales channels, customer segments, delivery outcomes and customer ratings. This creates an opportunity to connect commercial performance with operational execution and customer experience rather than evaluating sales in isolation.

Recorded sales	Orders	Customers	Return + cancel rate
INR 2.19B	100,000	14,988	12.2%

2. Core business problem

Although the business generates substantial gross sales, leadership lacks a consolidated view of where value is created or lost across the customer journey. In particular, the organization needs to determine which combinations of category, geography, customer segment, channel, discount level and payment method are associated with strong sales, reliable delivery, high ratings and low post-order leakage.

The central decision problem is therefore:

How can the retailer prioritize profitable growth opportunities and operational improvements by identifying the drivers of sales, cancellations, returns, delivery outcomes and customer satisfaction?

3. Evidence from the current dataset

Area	Observed signal	Business implication
Order outcomes	Returned and cancelled records together represent 12.2% of records.	A material share of booked demand may not translate into retained revenue or satisfied customers.
Category mix	Electronics is the largest sales category; Tablet is the leading product by recorded sales.	Growth and inventory decisions should distinguish high-volume categories from high-risk categories.
Channel behavior	72.2% of records are marked Online.	Channel-level conversion, fulfillment and return behavior should be compared before reallocating marketing or operations investment.
Customer experience	Average rating is 4.04/5, with 9.2% missing ratings.	Ratings can guide service improvement, but missing feedback must be handled explicitly.
Data quality	Age is missing for 3.5% of records; status labels include case variants.	A clean analytical layer is required before KPI reporting or modeling.

4. Analytical objectives

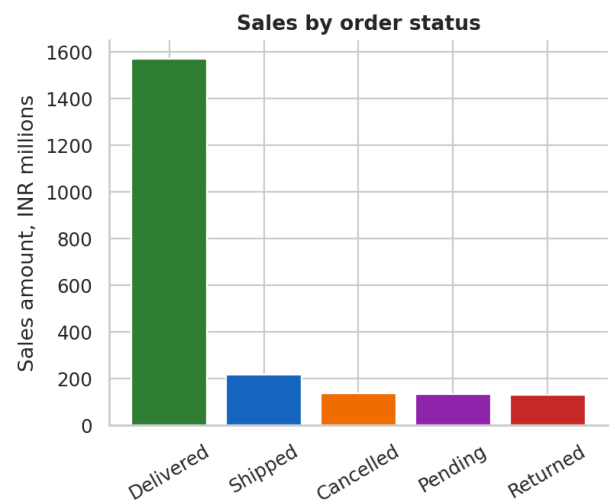
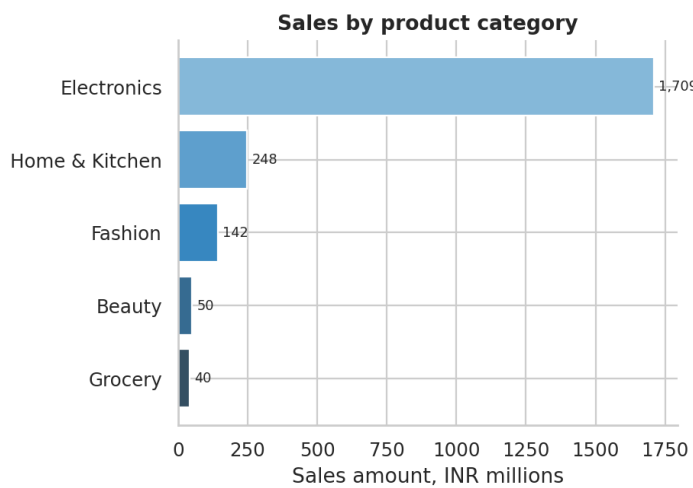
- **Quantify commercial performance** by category, product, state, city, customer segment, sales channel and payment method.
- **Diagnose revenue leakage** by identifying the drivers and concentration of cancelled and returned orders.
- **Understand customer experience** by linking ratings and delivery outcomes to products, geographies and customer segments.
- **Evaluate discount effectiveness** by testing whether higher discounts are associated with incremental sales or higher failure rates.
- **Improve decision prioritization** with a ranked list of growth, retention and operational improvement opportunities.

5. Key business questions

- Which categories and products generate the most sales, and which have the highest return or cancellation exposure?
- Which states and cities contribute the most value after accounting for order outcomes and customer ratings?
- Does the Online channel outperform Store sales on revenue, average order value, delivery success and customer satisfaction?
- How do Consumer, Corporate and Home Office customers differ in spending, discount sensitivity, ratings and post-order behavior?
- Which payment methods are associated with cancellations, pending orders or lower ratings?
- Are there seasonal or monthly patterns in sales and delivery outcomes that can improve planning?
- Can orders be scored for return or cancellation risk early enough to support intervention?

6. Proposed analytical approach

Phase	Workstream	Output
1	Data preparation	Standardize dates and status labels, validate monetary fields, document missingness and create derived metrics such as net retained sales and average order value.
2	Descriptive analytics	Executive KPI view, category and geography rankings, channel and segment comparisons, monthly trend analysis.
3	Diagnostic analysis	Return/cancellation driver analysis, discount and payment-method assessment, rating and fulfillment relationship analysis.
4	Predictive opportunity	Optional risk model for cancellation/return propensity and customer/product opportunity scoring.
5	Decision support	Prioritized recommendations, dashboard specification and KPI ownership framework.



7. Recommended KPI framework

KPI	Definition	Decision use
Gross sales	Sum of recorded Sales_Amount	Track demand and category/channel scale.
Retained sales	Sales excluding cancelled and returned orders	Measure realized commercial value.
Cancellation rate	Cancelled orders / total orders	Identify preventable demand loss.
Return rate	Returned orders / delivered or completed orders	Monitor product and experience quality.
Average order value	Sales divided by unique orders	Support customer and basket growth actions.
Delivery success rate	Delivered orders / total orders	Monitor operational reliability.
Average rating	Mean customer rating	Track perceived experience and product fit.
Discount efficiency	Incremental or retained sales relative to discount cost	Avoid unprofitable promotion strategies.



8. Scope and assumptions

This problem statement is designed for the supplied transaction-level dataset. It assumes that Sales_Amount is the recorded order value and that Delivery_Status is the available proxy for order completion. Profit, shipping cost, inventory availability, marketing spend, delivery time and customer lifetime history are not present in the file and should be added before making final profitability or causal claims.

The first analytical release should focus on descriptive and diagnostic insights. Predictive modeling should follow only after status definitions, retained-revenue logic and business ownership of the KPIs have been agreed.

9. Definition of success

The initiative will be successful if it produces a trusted KPI layer and actionable findings that enable leadership to identify the highest-value growth segments, reduce avoidable cancellations and returns, improve delivery and customer experience, and make promotion and channel decisions using retained value rather than gross sales alone.

Suggested first deliverable

An executive dashboard and analytical workbook covering sales, retained sales, order outcomes, category/product performance, geography, customer segment, channel, discounts, payment methods and ratings, supported by a short list of prioritized interventions with measurable owners and targets.